STUDENT LABORATORY MANUAL

Module 6: Introduction to Computer Graphics and Visual Computing

20 Guided Examples • 20 Practice Exercises • 20 Outputs

Course: CS0045 – Computer Graphics and Visual Computing

Companion: Module 6 Student Guide

Toolchain: VS Code + MSYS2 UCRT64 GCC + GLEW + FreeGLUT

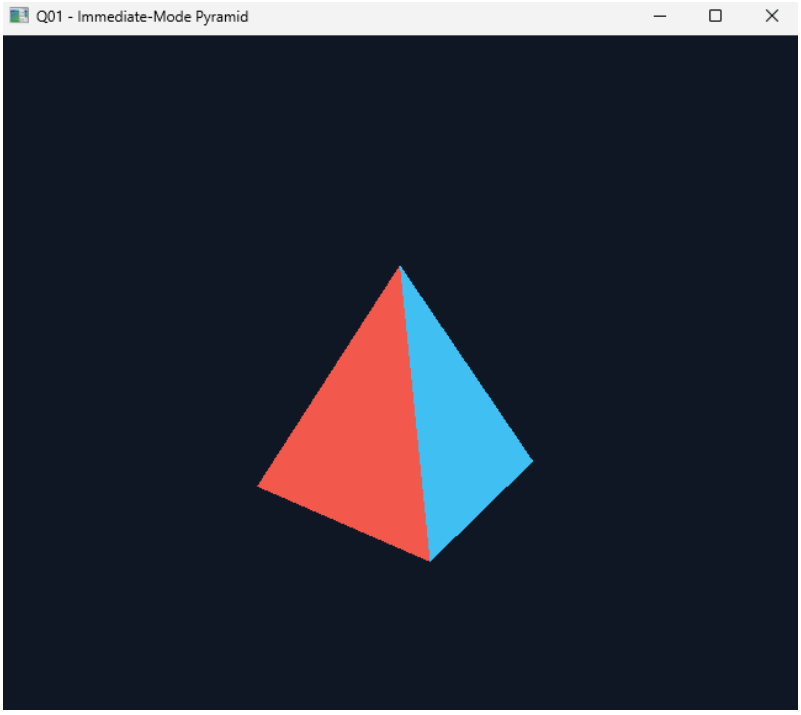
Student Name:	**Ebron, Nicole Anne V.**
Section:	**TN35**
Professor:	**Dr. Hadji J. Tejuco**

GitHub Repository: https://github.com/nicoleanneebron-art/Ebron-Nicole-Anne-V.

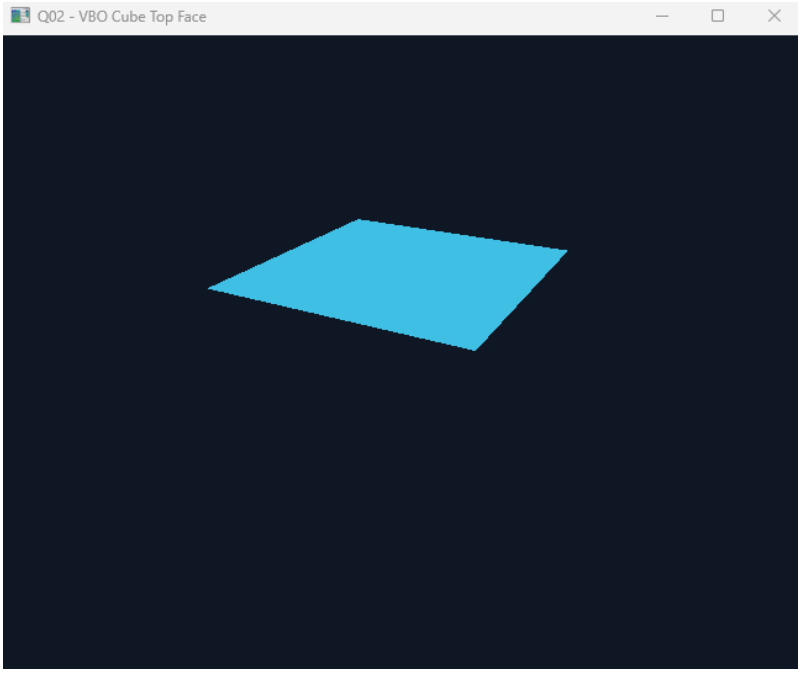
Part C — Outputs / Screen Shots (Ex01–Ex20)

Easy Exercises

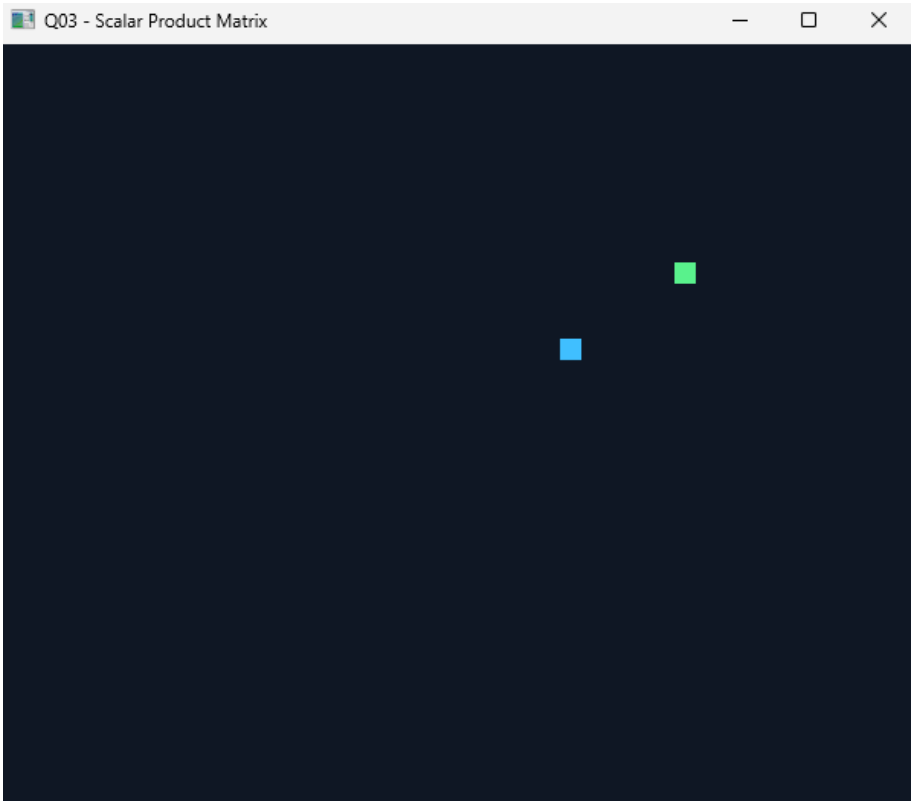
Exercise Q01 — Immediate-Mode 4-Sided Pyramid [Easy]



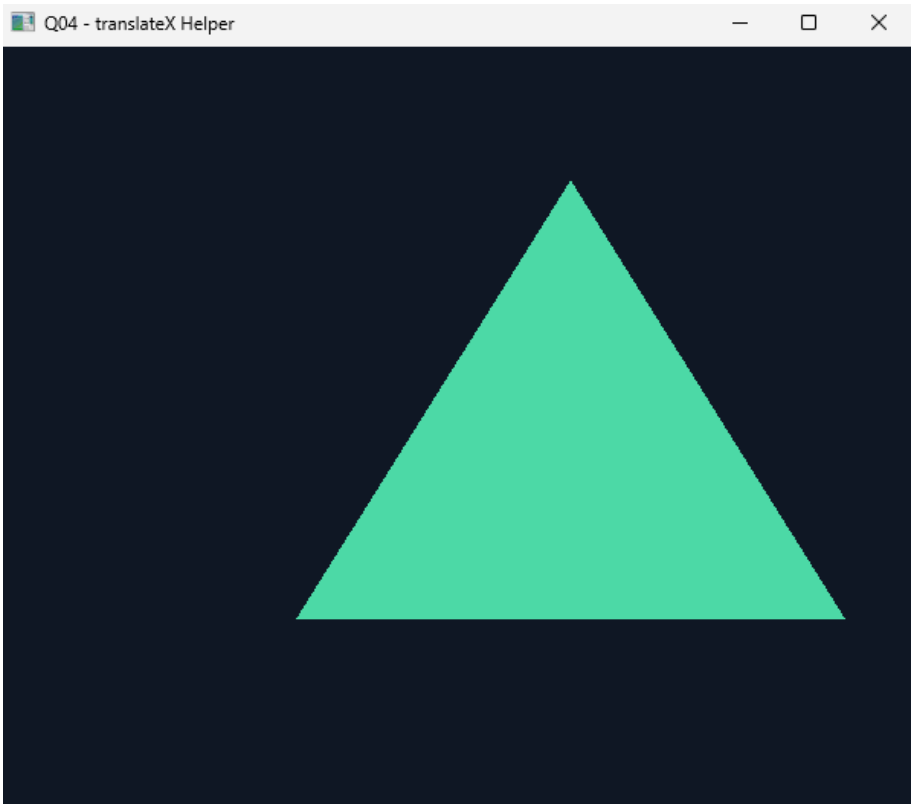
Exercise Q02 — VBO Conversion of One Cube Face [Easy]



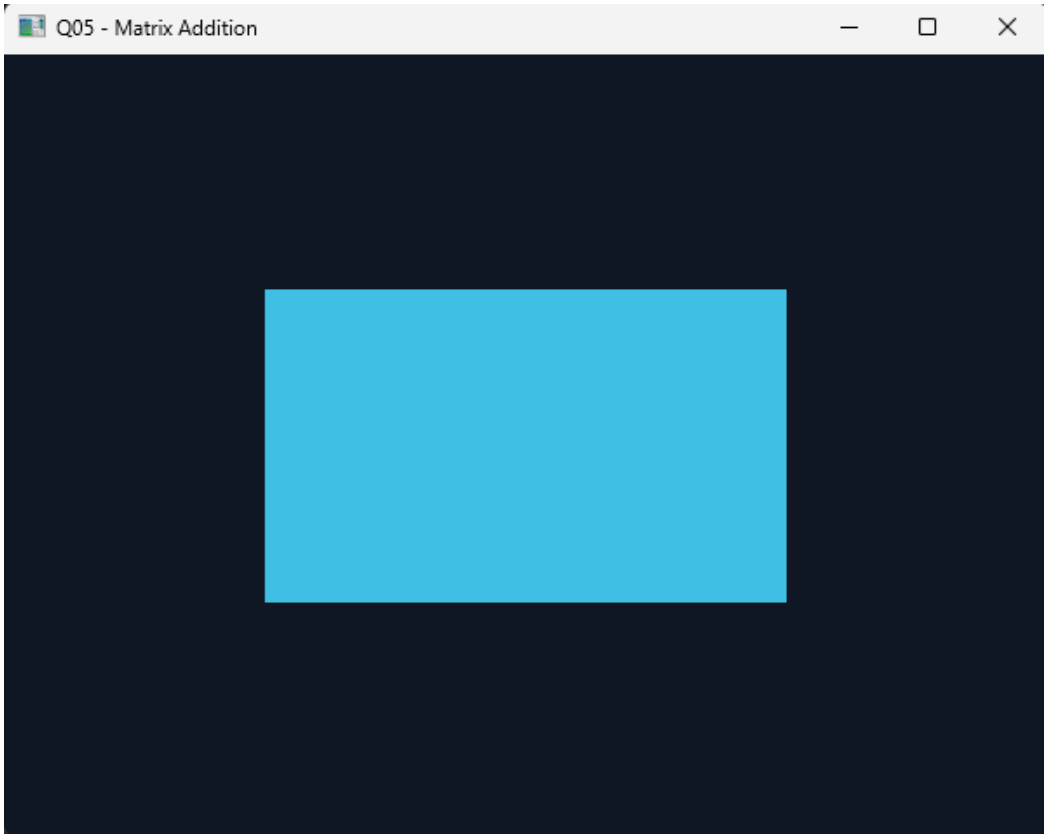
Exercise Q03 — Scalar Product Matrix, By Hand [Easy]



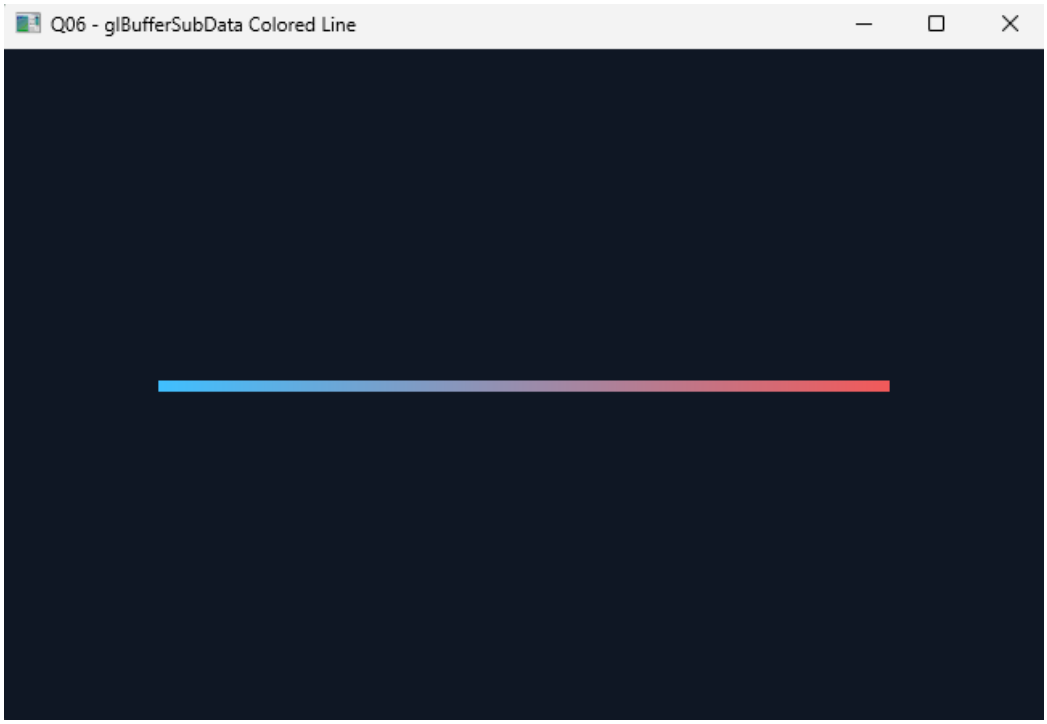
Exercise Q04 — A Simple translateX() Helper [Easy]



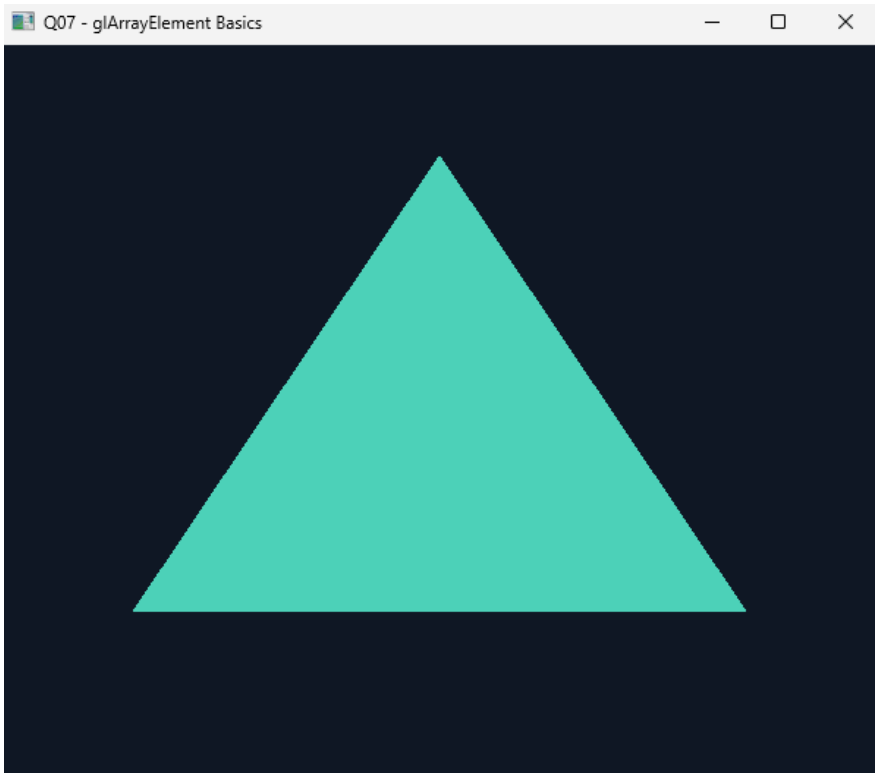
Exercise Q05 — Matrix Addition, By Hand [Easy]



Exercise Q06 — glBufferSubData for a Colored Line [Easy]

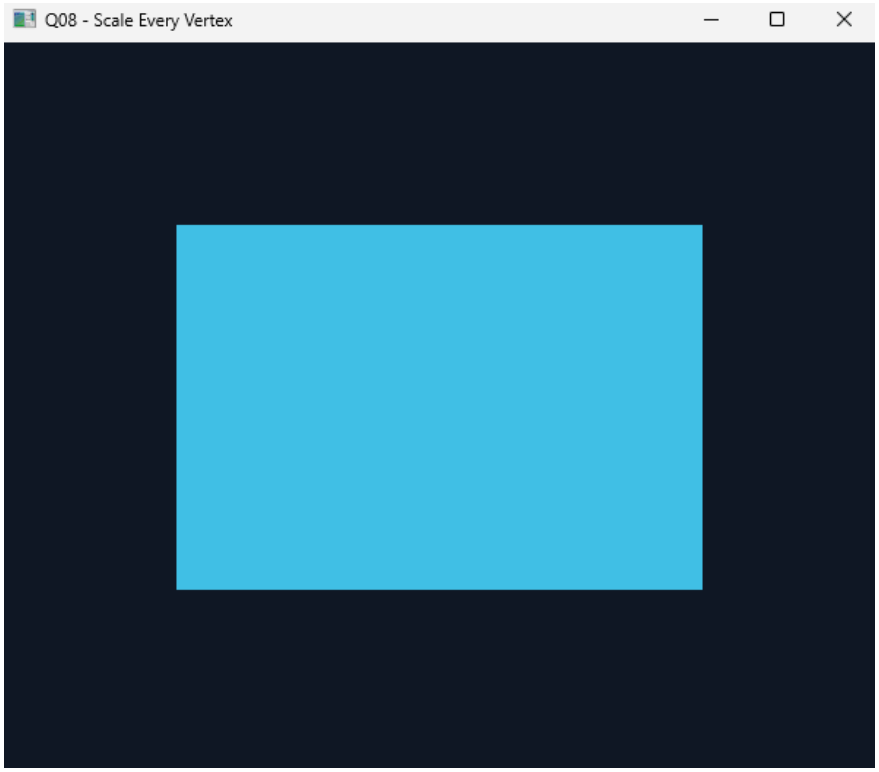


Exercise Q07 — glArrayElement Basics [Easy]

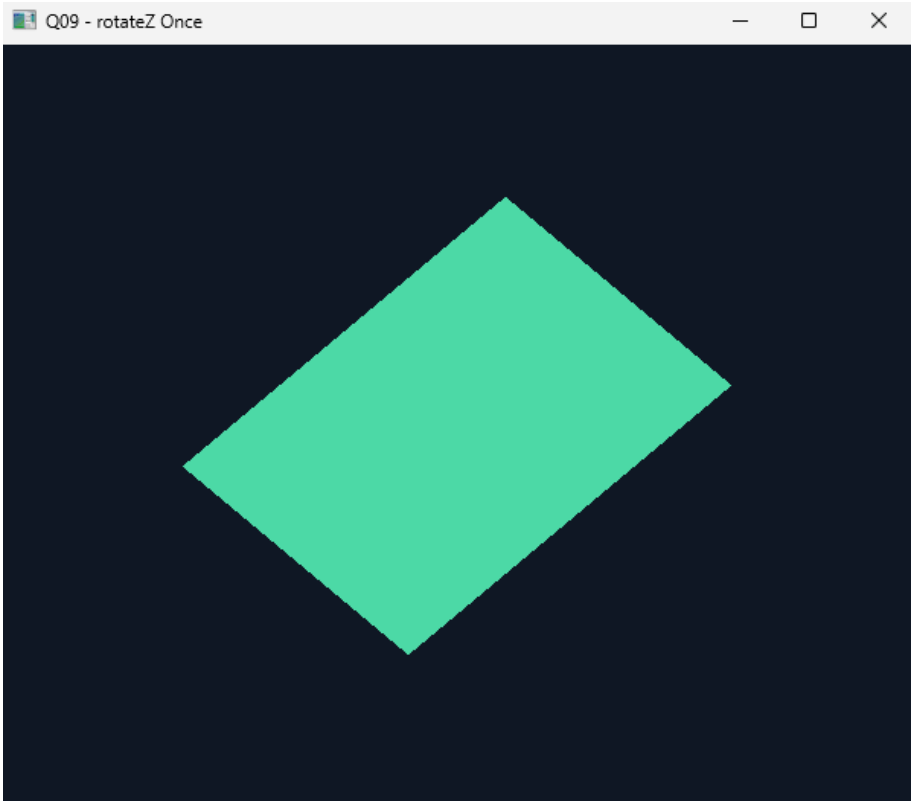


Medium Exercises

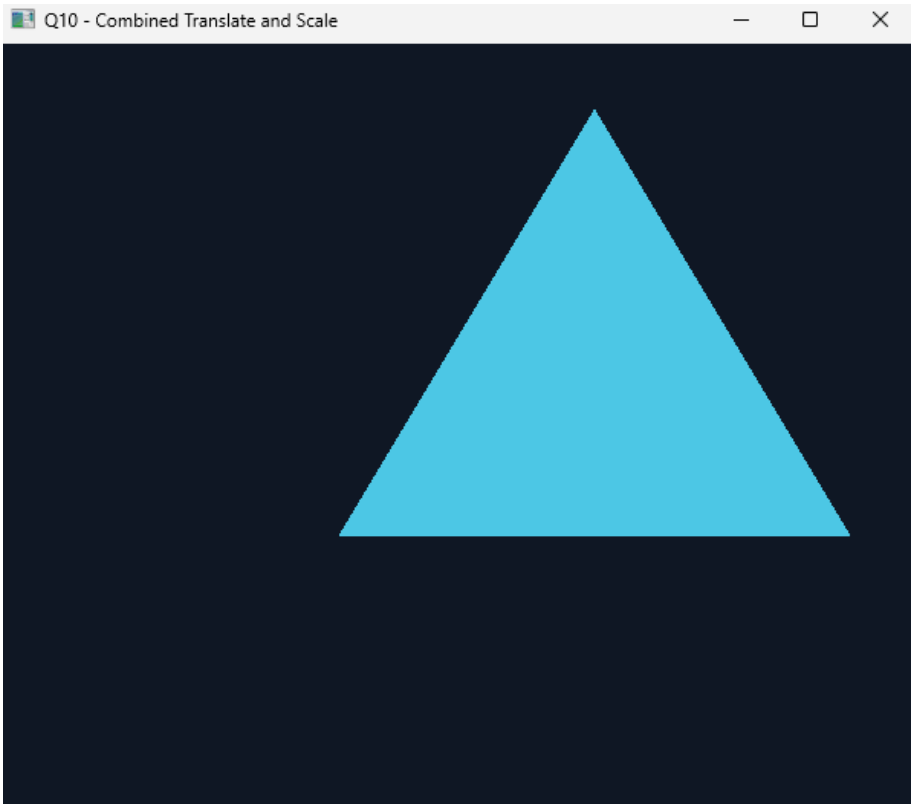
Exercise Q08 — scale() Applied to Every Vertex of a Shape [Medium]



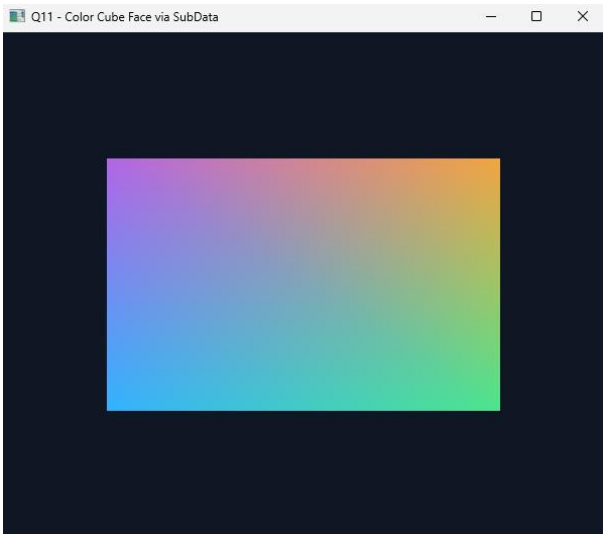
Exercise Q09 — rotateZ() Applied Once [Medium]



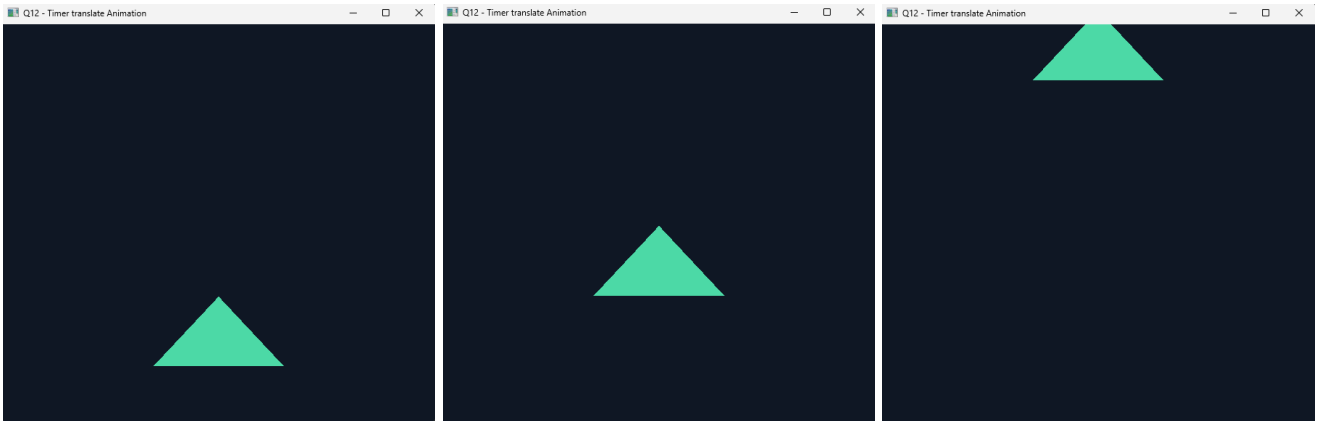
Exercise Q10 — Combining Translate and Scale [Medium]



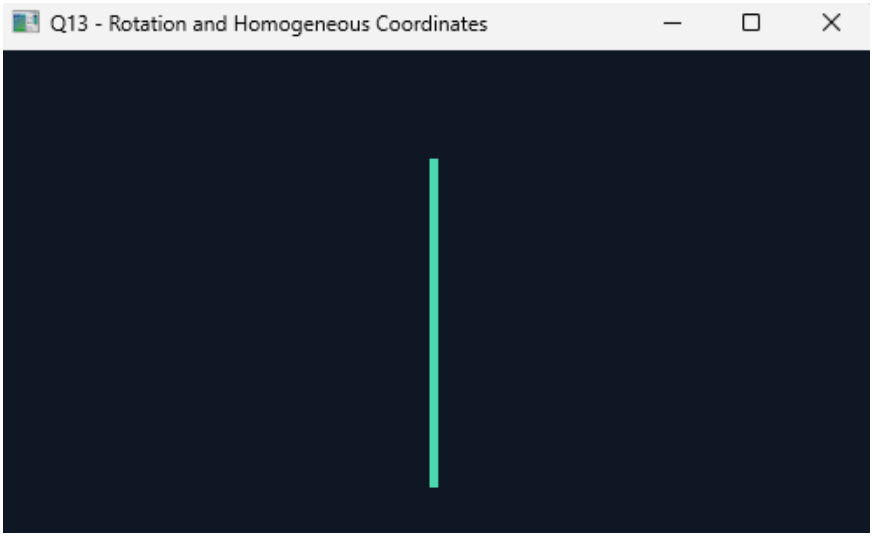
Exercise Q11 — Color Cube Face via glBufferSubData [Medium]



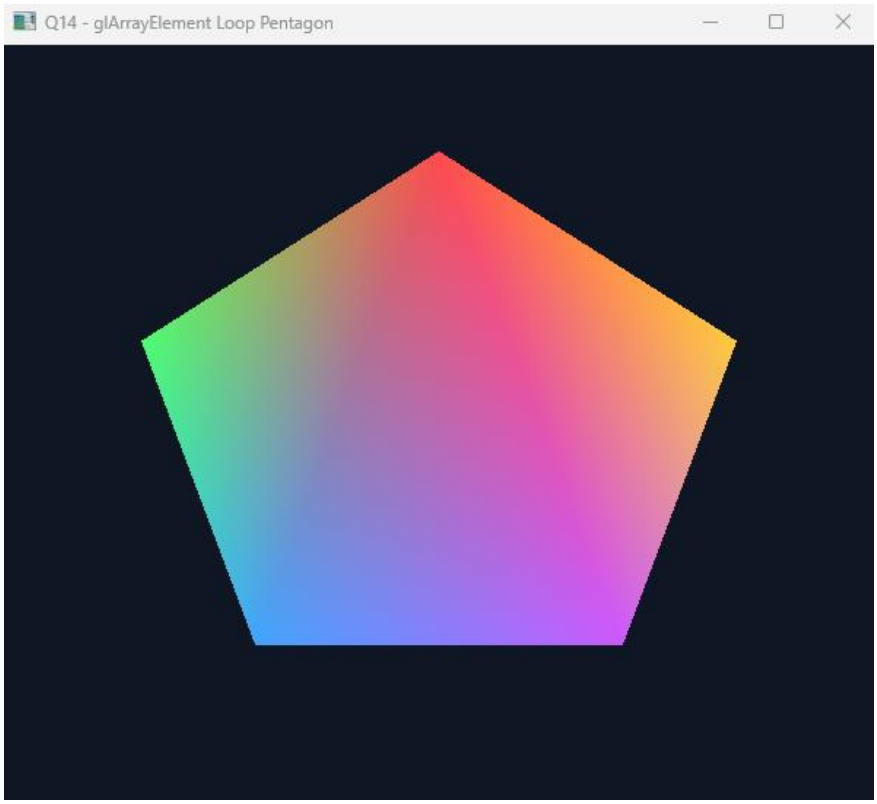
Exercise Q12 — Timer-Driven translate() Animation [Medium]



Exercise Q13 — Rotation and Homogeneous Coordinates [Medium]

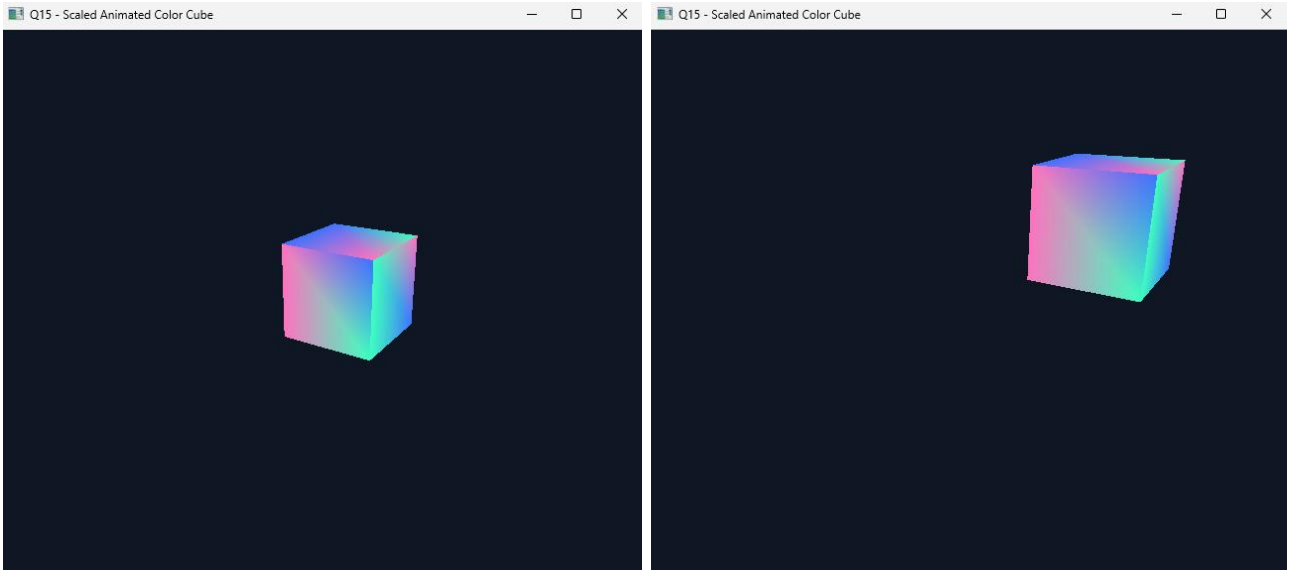


Exercise Q14 — glArrayElement in a Loop, Five Colors [Medium]

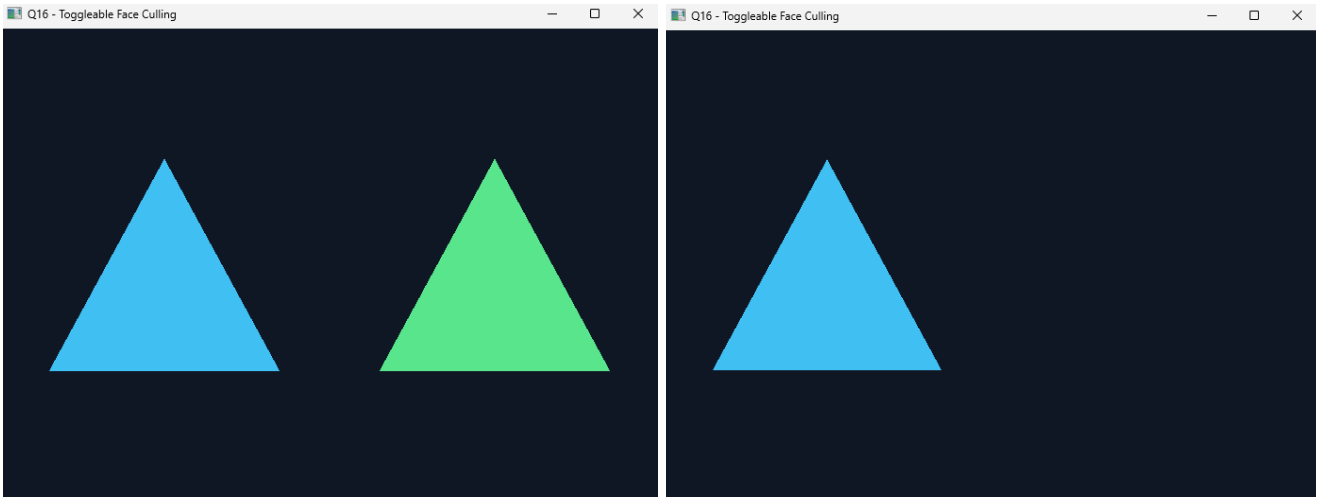


Hard Exercises

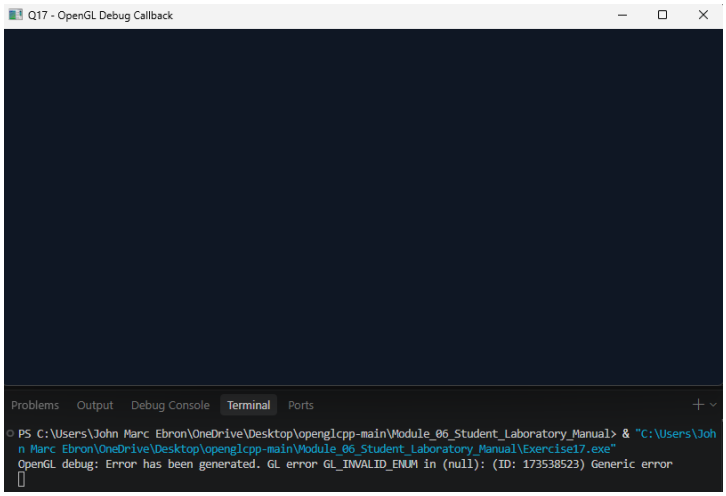
Exercise Q15 — Scaled-Then-Animated Color Cube [Hard]



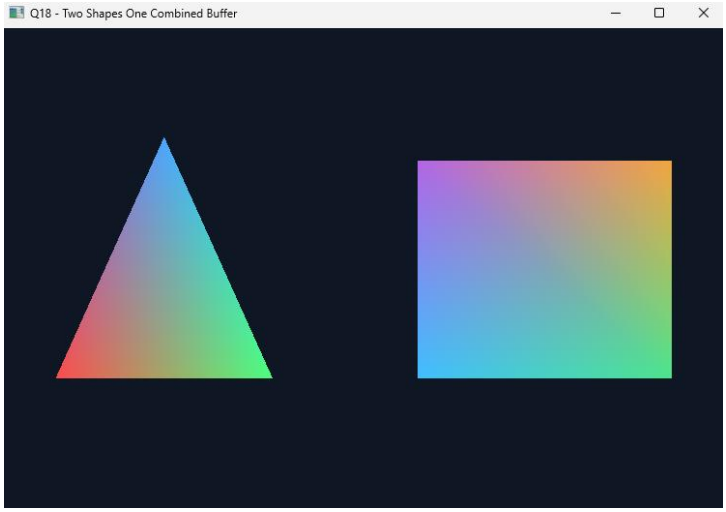
Exercise Q16 — Toggleable Face Culling [Hard]



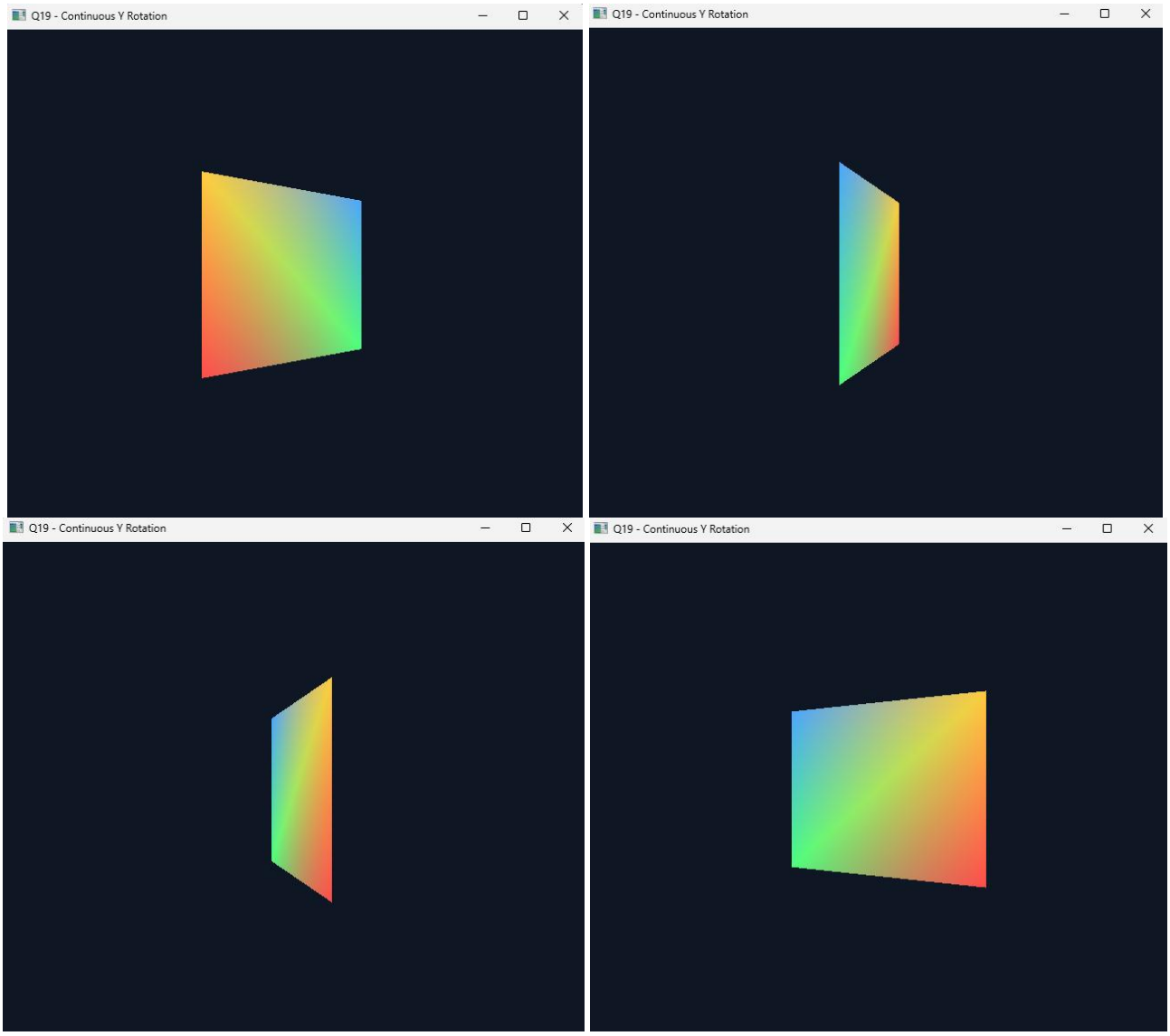
Exercise Q17 — Triggering the Error Callback [Hard]



Exercise Q18 — Two Shapes, One Combined Buffer [Hard]



Exercise Q19 — Continuous Y-Axis Rotation With Cleanup [Hard]



Exercise Q20 — Capstone: Rotating and Drifting Color Cube [Hard]

